MEE 342 - Course Project
Phase 3: Prototyping, Testing, and Reflection

Torque About It

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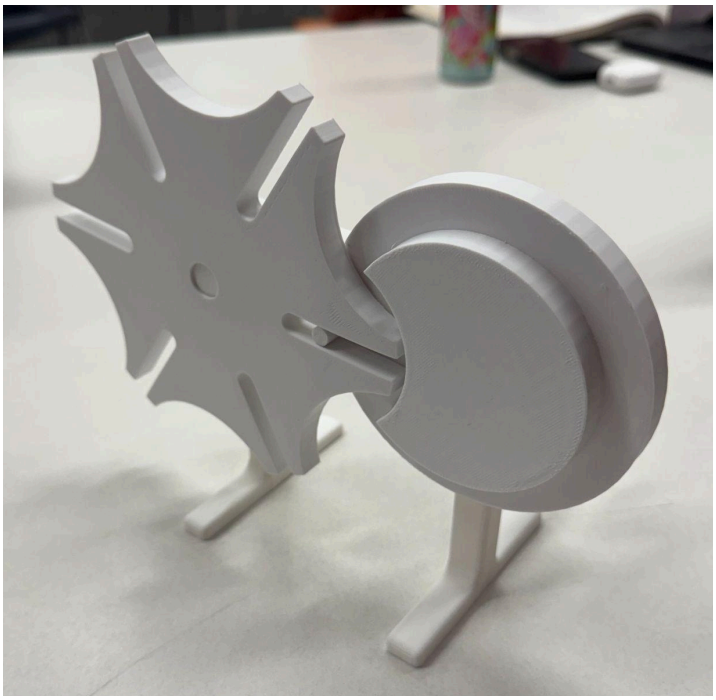

Fabrication Details

All components of the Geneva drive mechanism were exported as individual STL files and printed using an Ender-3 V3 SE FDM printer with a 0.4 mm nozzle. Parts were modeled in inches and exported with correct units to ensure proper scaling. PLA filament was used due to its ease of printing. A layer height of 0.2 mm and infill of approximately 40% were selected to balance print quality, strength, and time. Parts were oriented to minimize support material and improve surface quality on contact regions. Most components were printed flat, while the drive wheel was printed on its side so the pin was vertical, improving strength and eliminating supports. Design adjustments were made prior to printing to account for FDM tolerances. Hole diameters were increased (e.g., from 0.38 in to ~0.40 in), and slot dimensions were adjusted to ensure proper fit. Minor inaccuracies such as undersized holes and surface roughness were expected and considered.

Assembly Procedure

The Geneva drive mechanism was assembled by combining the printed components in sequence to ensure proper alignment and motion. The main shaft was first inserted into the support frame to establish the primary axis of rotation, followed by mounting the Geneva wheel onto the shaft so it could rotate freely without interference. The drive wheel was then attached to the input shaft, with the drive pin aligned to engage the slots of the Geneva wheel. The crank handle was

connected to the input shaft to allow manual rotation of the mechanism. After assembly, the system was manually rotated to check for alignment, smooth motion, and proper indexing. Minor adjustments were made to improve fit, including ensuring all components were fully seated and that no unintended contact occurred between moving parts.

Issues Encountered During Assembly

Several issues were encountered during fabrication and assembly. The most significant occurred when printing the main support frame with the two vertical legs. The initial print failed due to the need for extensive support material, which caused poor print quality and instability. The part was successfully reprinted after adjusting orientation and using additional supports. During assembly, minor fit issues were observed. Some components required slight force to assemble due to dimensional inaccuracies from the FDM process, such as undersized holes and surface roughness. Certain joints also required adhesive, indicating that connection methods were not fully considered in the original design. Friction between moving parts was another issue, especially between the drive pin and Geneva slots, causing slight resistance during operation. Bearings were added to improve alignment and reduce friction. Overall, the mechanism functioned as intended, but these issues emphasized the importance of print orientation, tolerances, and assembly considerations.

Testing and Results

The assembled Geneva drive was tested by manually rotating the input crank and observing the motion of the Geneva wheel. A specific test was performed by rotating the input one full revolution and measuring the resulting angular displacement of the Geneva wheel. The expected result was a 60° rotation per input revolution for the six-slot design. This behavior was observed during testing, as each full rotation consistently advanced the wheel by approximately one-sixth of a turn. The mechanism operated as intended, with proper engagement and disengagement of the drive pin. However, slight resistance was noticed during engagement, likely due to surface roughness and minor dimensional inaccuracies from 3D printing. The addition of bearings improved alignment and reduced friction, resulting in smoother motion. Overall, the mechanism demonstrated reliable and repeatable indexing with only minor performance limitations.

Comparison to Phase 2 Predictions

In Phase 2, the Geneva drive was modeled in CAD, where the motion study predicted smooth and consistent indexing with no interference or resistance. The simulation showed that each full rotation of the drive wheel would produce a precise angular displacement of the Geneva wheel. In the physical prototype, the overall motion closely matched these predictions. The mechanism produced consistent 60° indexing steps, confirming that the kinematic design was correct and that the pin-slot engagement functioned as intended. However, differences were observed between the predicted and actual performance. The CAD model assumed ideal conditions, while

the prototype exhibited slight resistance due to surface roughness, dimensional inaccuracies from 3D printing, and minor misalignment. The need for glue and added bearings also showed that assembly conditions were not fully captured in the model. Overall, the kinematic behavior matched Phase 2 predictions, but the prototype highlighted practical factors such as friction, tolerances, and assembly constraints that were not present in the simulation.

Reflection and Design Improvements

The Phase 3 prototyping process revealed several limitations in the original design and assumptions made during Phase 2. The CAD model assumed ideal geometry and perfect fits, while the physical prototype showed dimensional inaccuracies from 3D printing, leading to tighter fits and increased friction, especially between the drive pin and Geneva slots. Assembly considerations were also limited in the original design. Some components required adhesive, indicating that connection methods were not fully accounted for. The addition of bearings after assembly further showed that alignment and friction reduction were not sufficiently addressed. The printing process also highlighted manufacturability concerns, as the support frame required multiple attempts due to excessive support material. For a second iteration, tolerances would be increased to better account for printing variability and reduce friction. Assembly features such as press fits or fasteners would be added to eliminate the need for adhesive. The design would also better integrate bearings from the start, and part geometry and orientation would be adjusted to reduce support requirements. Overall, the prototype confirmed that while the design was functionally successful, manufacturability, tolerances, and assembly considerations are critical for achieving reliable performance.